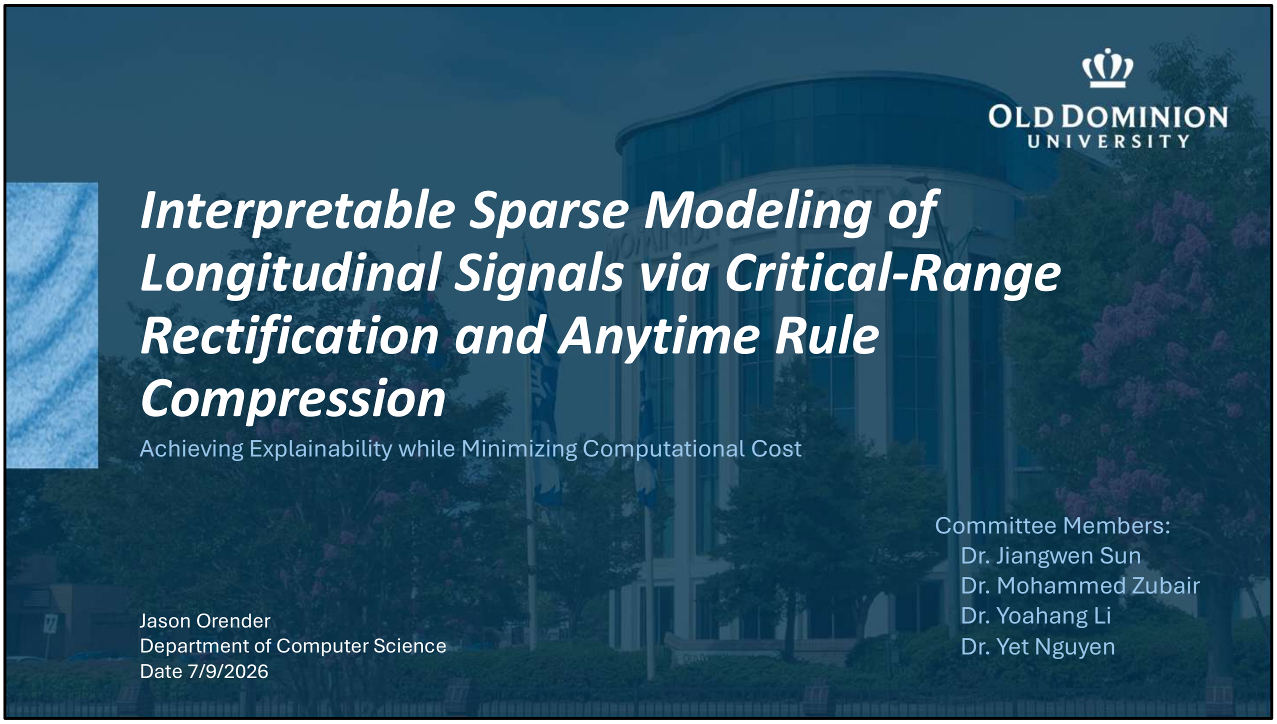


SCRIPT
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Good afternoon, and thank you for being here.
My dissertation is titled **"Interpretable Sparse Modeling of Longitudinal Signals via Critical-Range Rectification and Anytime Rule Compression."**
The central problem I address is how to build models for longitudinal signals that are not only predictive, but also interpretable, sparse, computationally practical, and usable in settings where humans need to understand the basis for a decision.
The work develops an end-to-end framework that begins with raw longitudinal sensor-style data, transforms it through critical-range rectification, fits a sparse model, and then optionally compresses that model into compact logical rules when doing so is justified by held-out validation.

NOTES
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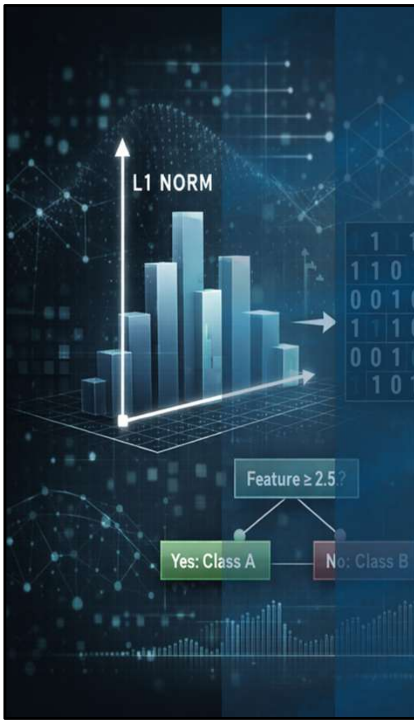
Many longitudinal problems are driven by threshold-and-lag behavior, but continuous high-dimensional signals are strongly correlated and make sparse selection unstable. I show that **critical-range binarization/rectification** improves recoverability and interpretability for L1 models in a variety of settings and applications, then I show that I

can compress the resulting sparse models in many cases into **compact rule lists** with an **anytime** procedure, finally delivering the full workflow as a reproducible python package (**CUTLASS**).




# Overview

**Thesis:**

Critical-range rectification improves the reliability and interpretability of sparse longitudinal modeling; anytime rule compression makes the result operationally usable.

**01. Problem and Pipeline**

Introduce the core problem, knowledge gap, and pipeline.

**02. Answers to Research Questions**

RQ1: Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?
RQ2: Why does rectification work, and how far does the theoretical explanation generalize?
RQ3: Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?

**03. Boundaries and Implications**

Discuss limitations, scope boundaries, and practical implications.

**04. Contributions and Future Work**

Summarize the contributions, extensions, and closing implications.

SCRIPT
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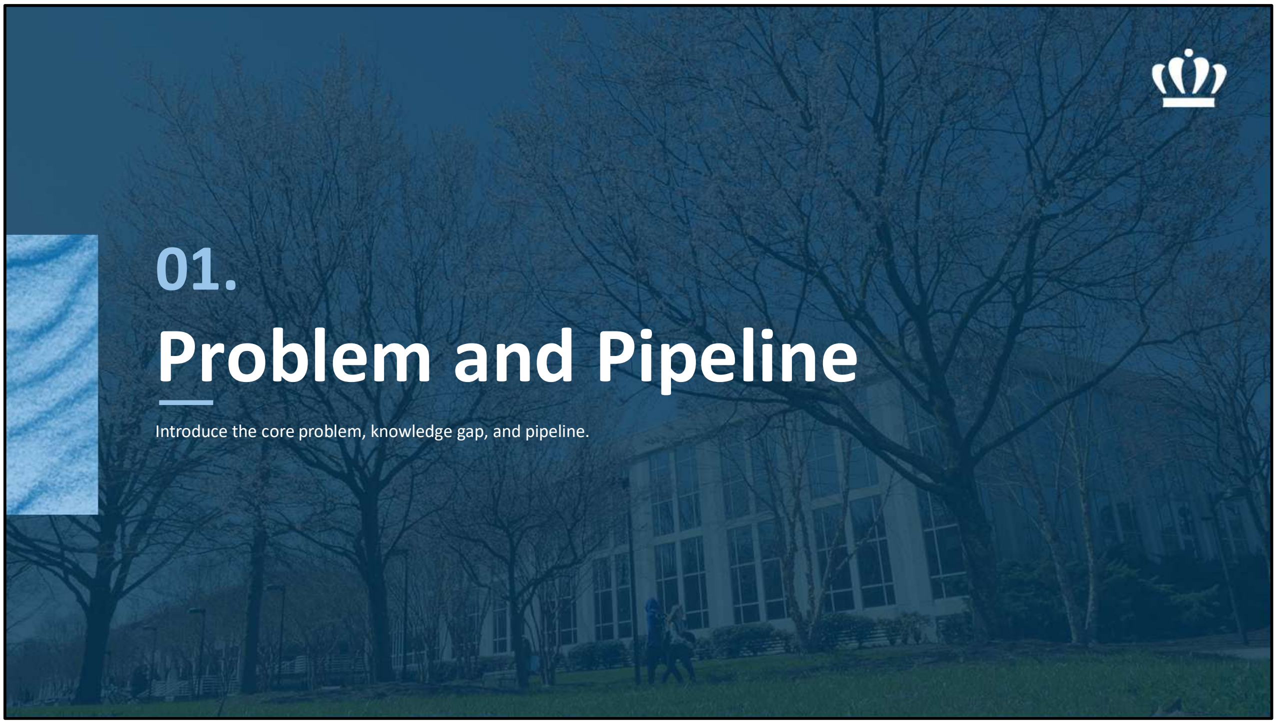
This slide gives us the roadmap for the defense.
The thesis is that **critical-range rectification improves the reliability and interpretability of sparse longitudinal modeling**, and that **anytime rule compression makes the result operationally usable**.
I will start with the problem and the knowledge gap. Then I will walk through the answers to the three research questions.
The first question is empirical: can rectification produce a stable sparse baseline with reliable feature and lag attribution?
The second question is theoretical: why does rectification work when it works, and how far does that explanation generalize?
The third question is operational: can the sparse solution be compressed into compact logical rules without unacceptable loss in discrimination?
After that, I will discuss scope boundaries and practical implications, and then close with the main dissertation contributions and future directions.




SCRIPT
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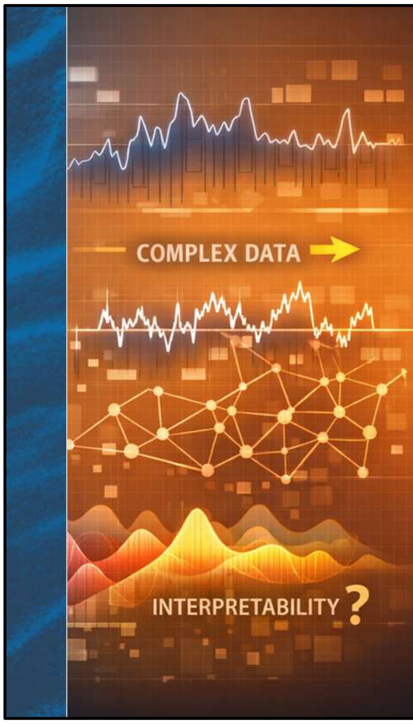
I will begin with the motivation for the work.
The goal of this first section is to establish the setting: longitudinal, high-dimensional, correlated data where the outcome is often driven by threshold and lag behavior.
This is also where I will explain why the dissertation takes a representation-first approach rather than relying only on changes to the sparse-learning penalty or using a post-hoc interpretability method.



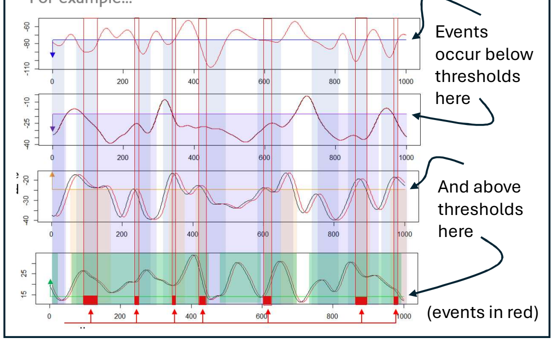

# The Problem

- Three tensions:
  - high dimension and correlation causing unstable support and weak lag attribution
  - operational compute and model-complexity constraints
  - high-stakes deployment requiring transparent and auditable rules

**Real World**

Longitudinal sensor, clinical, and industrial signals often reflect threshold behavior and lagged dependencies, complicating the creation of interpretable models.

For example...




SCRIPT
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The core problem is driven by three tensions.
First, high dimensionality and correlation create unstable sparse support. In longitudinal data, lag expansion means that neighboring time lags are often highly correlated. This makes it difficult for sparse models to select the right feature and the right lag reliably.
Second, operational settings often impose compute and model-complexity constraints. A model that is accurate but too large or too expensive may not be practical in some settings.
Third, high-stakes deployment often requires transparent and auditable rules rather than opaque scoring functions.
The details of the plots on the right are not important here. The point is that events can be associated with threshold regions at different lags, and those regions may differ by signal.
These three tensions motivate a representation-first pipeline for interpretable sparse longitudinal modeling.

NOTES

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- The details of the plots aren’t important here; the point is that events can be associated with threshold regions at different lags, and those regions may differ by signal.
- These three tensions motivate a representation-first pipeline for interpretable sparse longitudinal modeling.

## Gaps in Prior Work


### Accuracy, Interpretability, Sparsity, Efficiency

- Sparse penalties are mature but fragile under correlation and dependence.
- Structured/dependence-aware penalties often stay in raw correlated space.
- Rule learning is transparent but search-heavy with combinatorial scaling.
- Prior work lacked an end-to-end, reproducible synthesis across the method families reviewed.

SCRIPT
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The related work addresses parts of this problem, but not the full combination.
Sparse penalties are mature and computationally useful, but they can be fragile under correlation and dependence. They may predict well while selecting unstable supports.
Structured and dependence-aware penalties help in some settings, but they often continue to operate in the raw correlated feature space.
Rule-learning methods are transparent, but direct rule search can become search-heavy and combinatorial in high-dimensional longitudinal settings.
So the gap is not that prior methods fail completely. The gap is that the field lacks an end-to-end, reproducible synthesis across these method families: a pipeline that handles representation, sparse attribution, compression, and deployment decision logic together.

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The gap is not that none of these methods work. The gap is that they solve different parts of the problem separately: sparse selection, dependence control, interpretability, or rule learning. This dissertation combines those pieces into a representation-first, sparse, and policy-controlled pipeline.

# Cross-Family Positioning

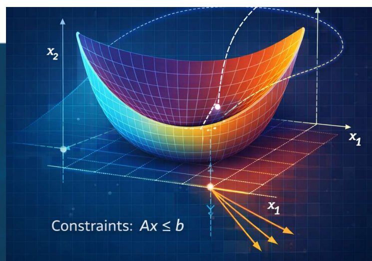
Several method families address parts of the problem identified in this dissertation, but each leaves a gap for interpretable, lag-aware, deployment-ready modeling.

### L1 / elastic-net / adaptive lasso

Changes representation before fitting, rather than relying only on penalty tuning in the original dependent feature space.

### Grouped / ordered sparsity

Targets threshold logic and lag semantics through feature redesign, not only coefficient constraints.




### Interpretable additive / rule-ensemble models (EBM / RuleFit)

The dissertation pipeline produces a shared rectified sparse baseline and, when justified, a smaller policy-screened rule artifact.

### Dependence-aware / Quadratic Programming / robust selectors

Couples redundancy reduction to explicit threshold semantics and a downstream rule path.

### Direct rule-list / LAD-style methods

Restricts logic compression to active support rather than searching the full lag-expanded space.

SCRIPT
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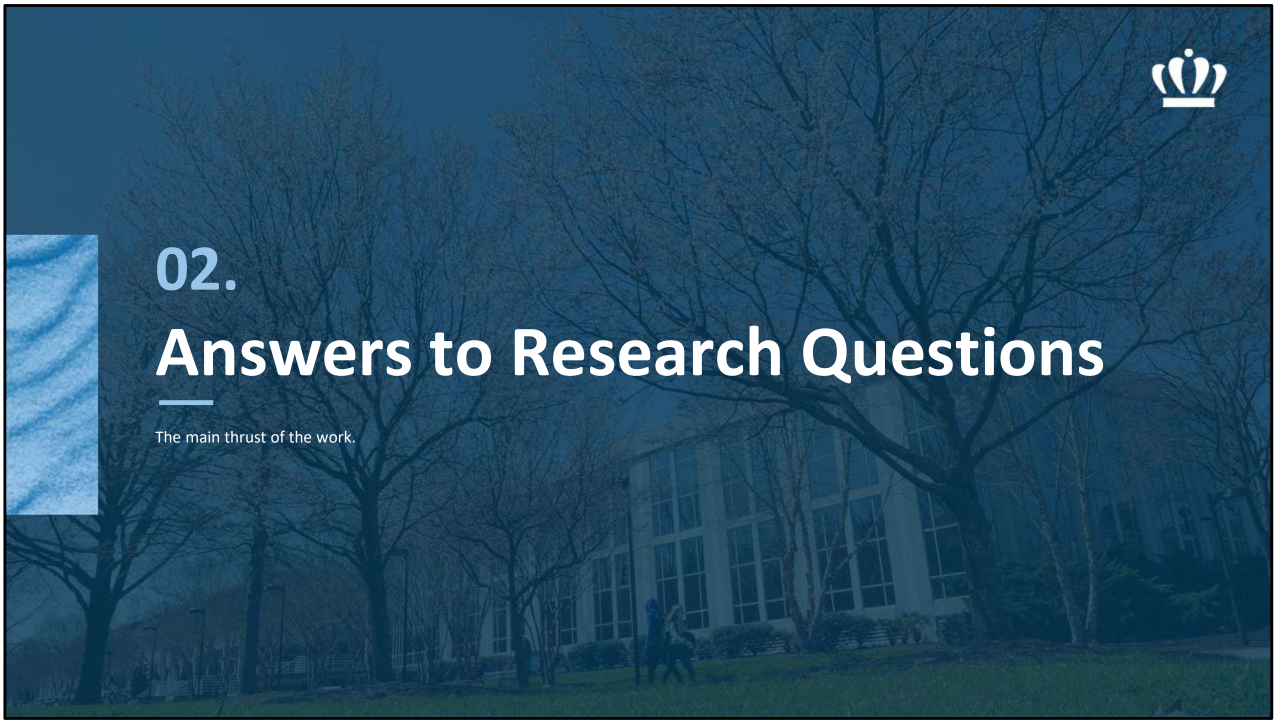
This slide positions the dissertation pipeline relative to the main competing method families.
Compared with L1, elastic net, and adaptive lasso, the key difference is that this work changes the representation before fitting, rather than relying only on penalty tuning in the original dependent feature space.
Compared with grouped and ordered sparsity, the pipeline targets threshold logic and lag semantics through feature redesign, not only through coefficient constraints.
Compared with dependence-aware methods and quadratic programming selectors, rectification couples redundancy reduction to explicit threshold semantics and a downstream rule path.
Compared with EBM and RuleFit-style approaches, this pipeline produces a shared rectified sparse baseline and, when justified, a smaller policy-screened rule artifact.
And compared with direct rule-list or LAD-style approaches, this method restricts logic compression to the active sparse support rather than searching the full lag-expanded space from scratch.




SCRIPT

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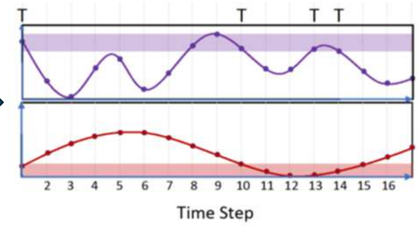
I will now move into the main body of the dissertation: the answers to the three research questions.

The sequence is intentional. RQ1 tests whether rectification improves sparse attribution in practice. RQ2 explains why that improvement can occur under a scoped theoretical mechanism. RQ3 asks whether the resulting sparse structure can be converted into a simpler rule artifact for operational use.

# Intuition Development

### Threshold and Lag Response Data
1

Suppose there is a situation in which sensor data from a data source predicts an event, but the event only occurs when the sensor readings are within a certain range and a specific amount of time ahead of the event (think: temperatures and pressures, to give a concrete example).




(events are demarcated by the “T”)

### Estimate Ranges In Which Events Occur
2

The training events can reveal the ranges in which events tend to occur.

### The Non-Obvious Point
3

Irrelevant continuous variation can obscure the true sparse support. Threshold-based rectification can suppress that variation while preserving the event-triggering structure.

### The Innovation

Creating a new train-only transformed representation based on how raw measurements relate to the modeled outcome can materially improve true-support recovery, attribution, and predictive modeling. RQ1 asks whether this representation change improves sparse feature-lag recovery in practice.

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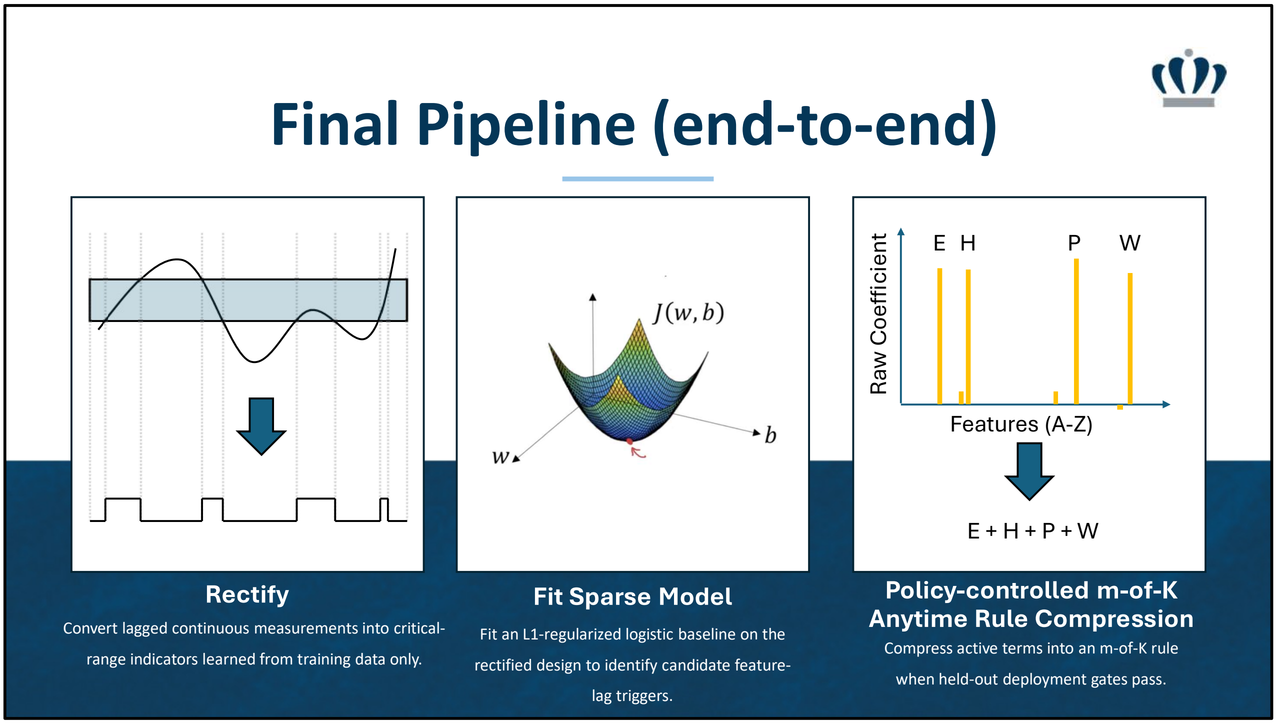


SCRIPT
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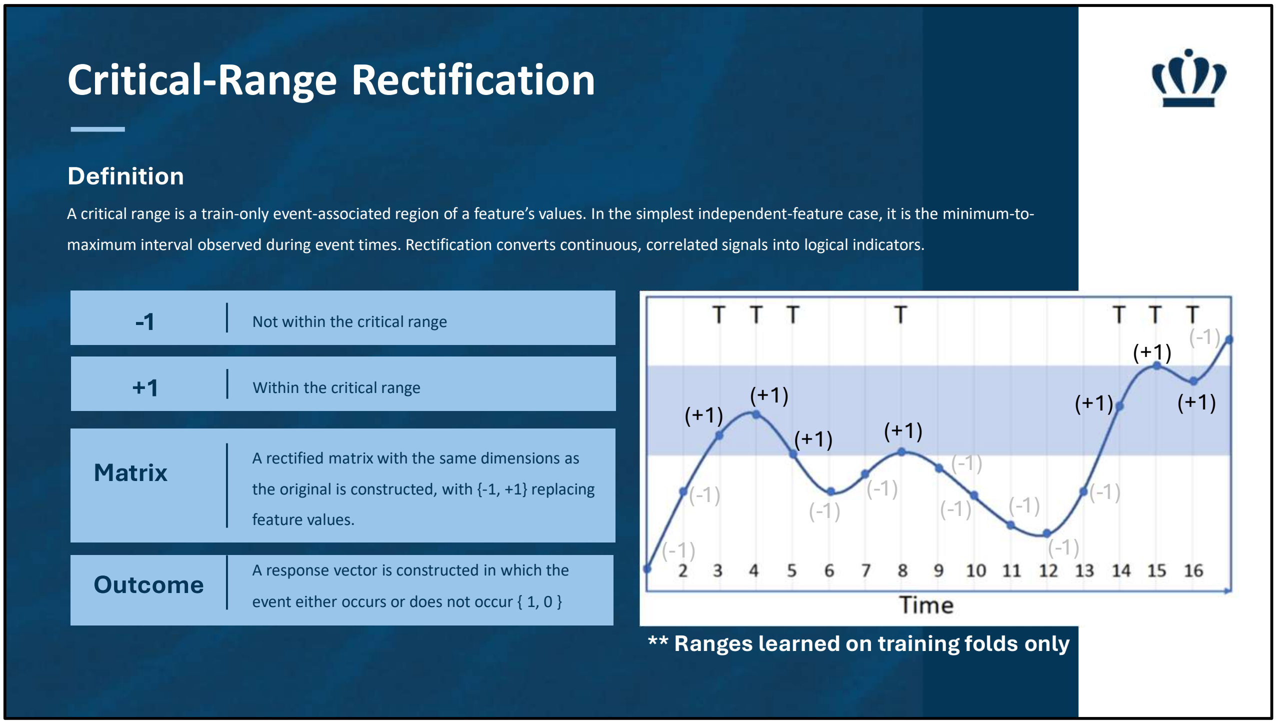
The final pipeline has three stages.
First, **rectification** converts lagged continuous measurements into critical-range indicators learned from training data only. The goal is to transform the representation before fitting, so that the model is operating on features that more directly encode threshold-event behavior.
Second, the method fits an **L1-regularized logistic baseline** on the rectified design. The sparse coefficient set identifies candidate feature-lag triggers.
Third, when a simpler artifact is needed, the active terms can be compressed into a **policy-controlled m-of-K anytime rule**. Compression is not automatic. It is accepted only when held-out deployment gates pass.
So the pipeline is not simply “transform, fit, deploy.” It is a branchable workflow with explicit checks at each stage.




SCRIPT
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This slide defines the core representation.
A critical range is a train-only event-associated region of a feature’s values. In the simplest independent-feature case, it is the minimum-to-maximum interval observed during event-positive training times.
The original continuous feature is then converted into a logical indicator: +1 when the measurement is within the critical range, and -1 when it is outside the critical range.
The important methodological safeguard is shown at the bottom: **ranges are learned on training folds only**. That prevents the test data from contaminating the representation.
This slide describes the empirical pipeline. Later, in the theory section, I narrow to zero-threshold sign binarization because that special case is analytically tractable.

NOTES
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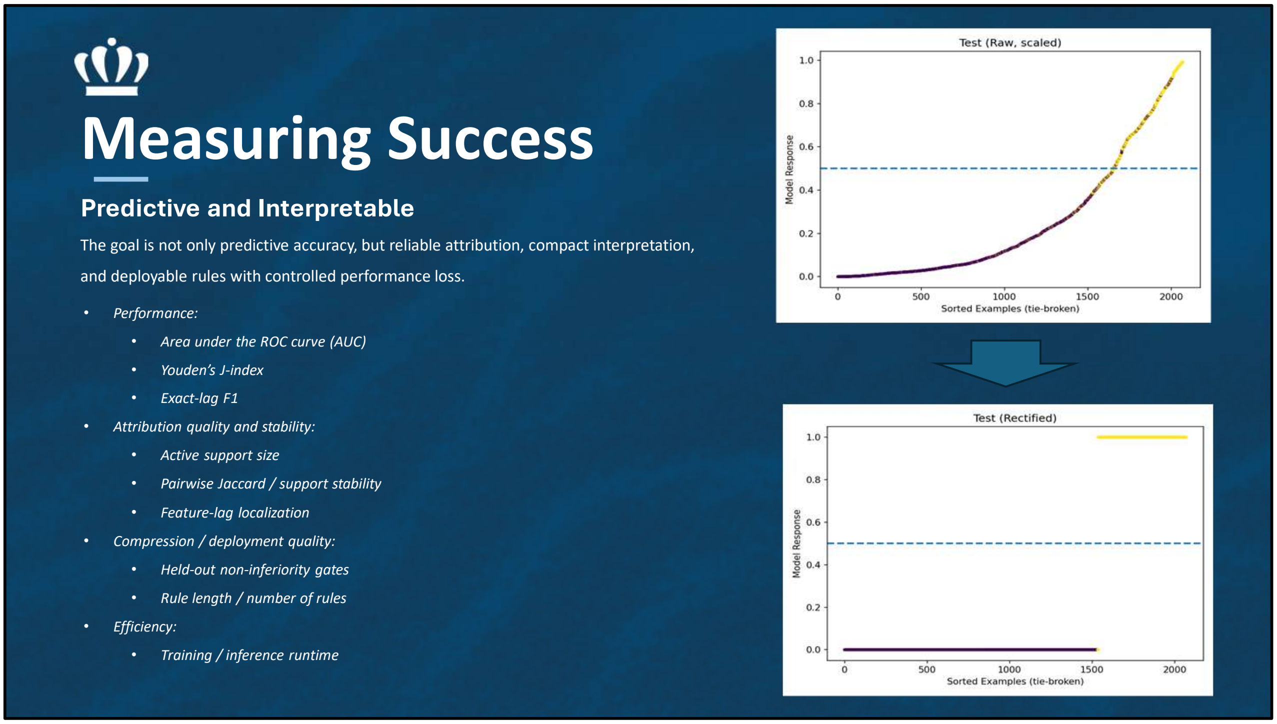
- Practical rectification can be one-sided or interval-like, while zero-threshold sign binarization is the theorem-scoped RQ2 special case.
- This slide describes the empirical pipeline. Later, the theory section narrows to zero-threshold sign binarization because that special case is analytically tractable.




SCRIPT
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The goal is not only predictive accuracy.
The dissertation evaluates four kinds of success.
The first is performance: AUC, Youden's J-index, and exact-lag F1.
The second is attribution quality and stability: active support size, pairwise Jaccard support stability, and feature-lag localization.
The third is compression and deployment quality: held-out non-inferiority gates, rule length, and number of rules.
The fourth is efficiency: training and inference runtime.
The plots on the right are illustrative. The point is that rectification and compression should not only preserve discrimination; they should also make the model more interpretable, more stable, and more usable.

NOTES
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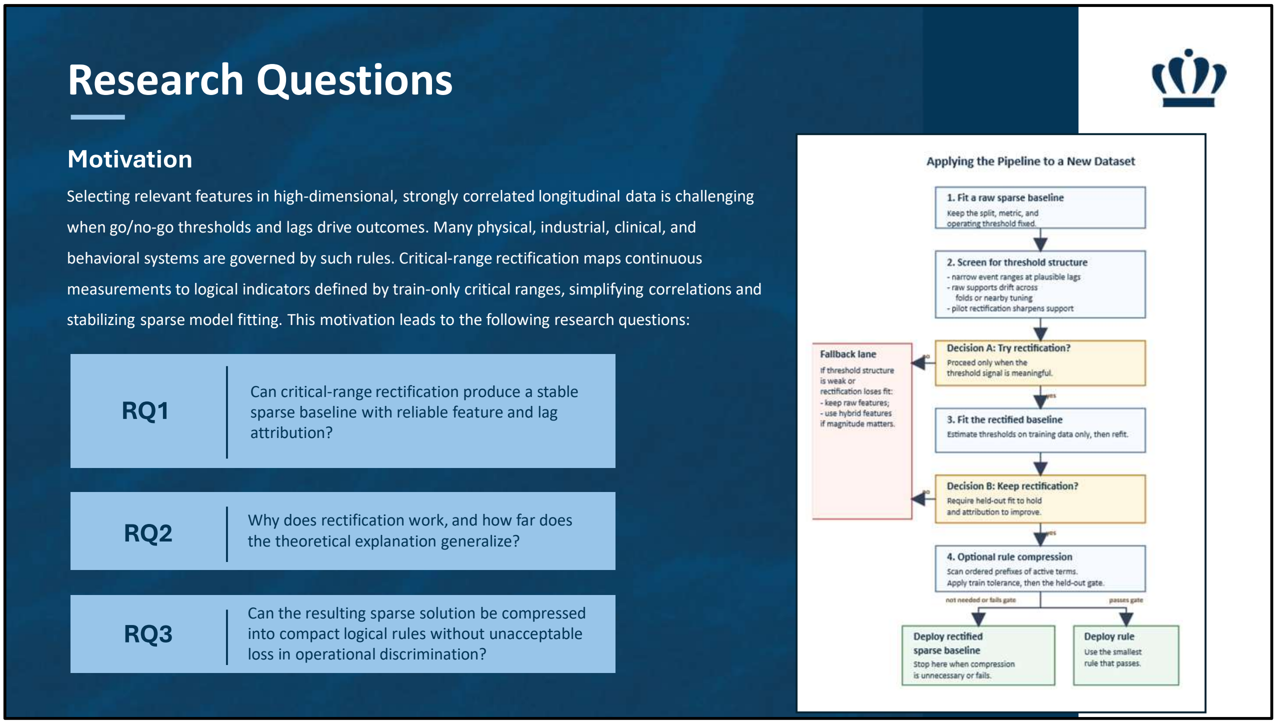
*Youden's J-index is used for its utility with imbalanced data sets*




SCRIPT

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This slide restates the three research questions in the context of the full workflow. The motivation is that selecting relevant features in high-dimensional, strongly correlated longitudinal data is challenging when thresholds and lags drive outcomes. Critical-range rectification maps continuous measurements to logical indicators defined by train-only critical ranges. That can simplify correlation structure and stabilize sparse model fitting.

The three questions then follow naturally.

RQ1 asks whether critical-range rectification can produce a stable sparse baseline with reliable feature and lag attribution.

RQ2 asks why rectification works, and how far the theoretical explanation generalizes.

RQ3 asks whether the resulting sparse solution can be compressed into compact logical rules without unacceptable loss in operational discrimination.

The figure on the right is the operational playbook: raw baseline, threshold diagnostic, rectified baseline, optional compression, or fallback if the gates fail.

NOTES

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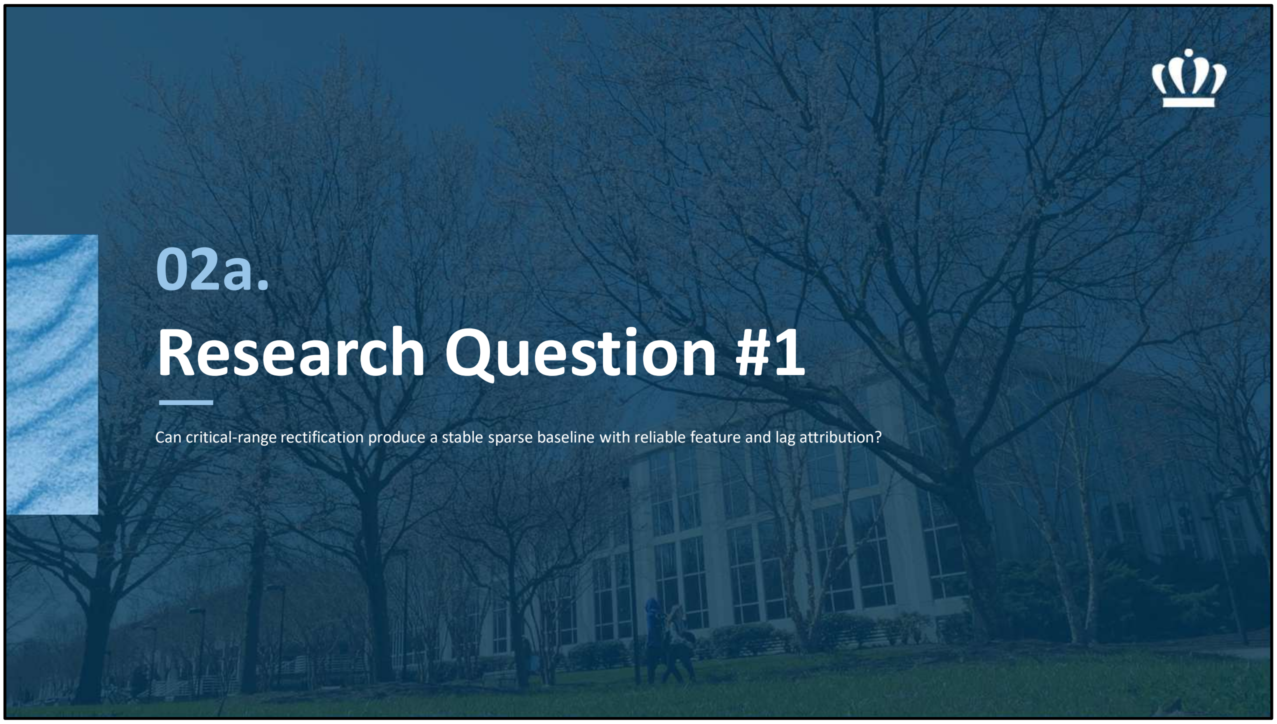
Verbally explain:  The right-side figure is the operational playbook: raw baseline, threshold diagnostic, rectified baseline, optional compression, or fallback if the gates fail.




SCRIPT
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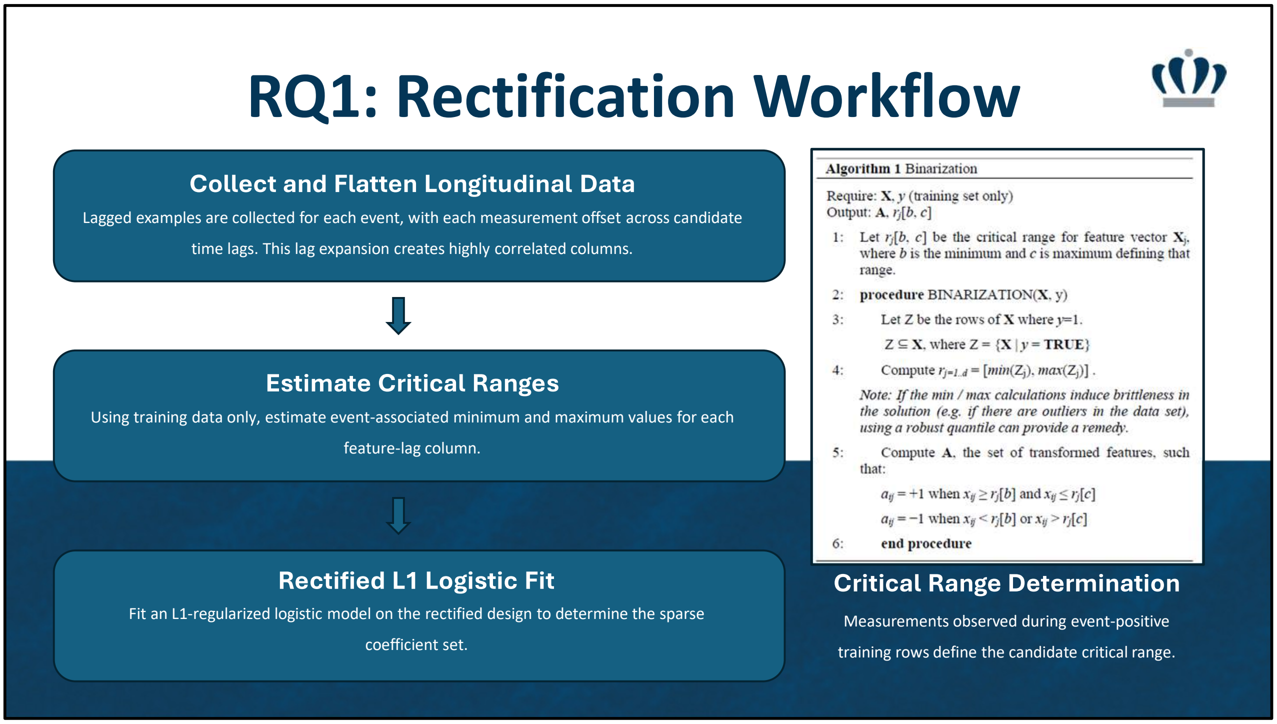
The first research question is:
**Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?**
This question is empirical. It asks whether the representation change actually improves sparse longitudinal selection in practice.
The evidence comes from synthetic data with known ground truth, comparative baseline studies, and real-world data.




SCRIPT

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This slide shows the RQ1 workflow.

First, longitudinal data are collected and flattened into lagged examples. Each measurement is offset across candidate time lags. This lag expansion is useful because it lets us model delayed effects, but it also creates many highly correlated columns.

Second, critical ranges are estimated using training data only. For each feature-lag column, event-positive training rows define the candidate critical range.

Third, an L1-regularized logistic model is fit on the rectified design to determine the sparse coefficient set.

The algorithm on the right is the formal version of these three boxes: lag-expanded inputs, train-only critical ranges, and rectified L1-logistic fitting.

NOTES

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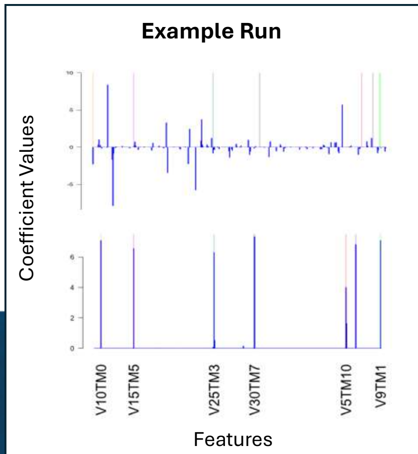
Verbally explain:  The algorithm on the right is the formal version of the three boxes: lag-expanded inputs, train-only critical ranges, then rectified L1-logistic fitting.

# RQ1: Synthetic Data




| Method | AUC | Youden's J-index | Exact-lag F1 | Pairwise Jaccard | Total Nonzero | Runtime (s) |
|---|---|---|---|---|---|---|
| Raw | 0.986 ± 0.008 | 0.882 ± 0.027 | **0.048** ± 0.082 | **0.056** ± 0.096 | **340.667** ± 105.396 | 23.142 ± 0.815 |
| Rectified | 0.998 ± 0.002 | 0.971 ± 0.019 | **0.619** ± 0.082 | **0.452** ± 0.090 | **190.000** ± 117.796 | 12.196 ± 1.562 |
| Rule | 0.994 ± 0.003 | 0.974 ± 0.015 | **0.759** ± 0.056 | **0.600** ± 0.346 | **6.000** ± 3.464 | 12.201 ± 1.562 |

**Compare Results across**
**Raw vs. Rectified vs. Rule Sparse Fits**

Across repeated synthetic resamples with known ground truth, rectification improved discrimination, support stability, exact-lag recovery, compactness, and runtime. The compressed rule preserved discrimination while reducing the active support to only a few terms.

SCRIPT
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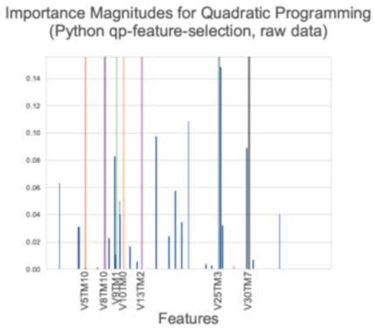
The synthetic data provide a controlled test because the ground-truth feature-lag structure is known.
The table compares raw sparse fitting, rectified sparse fitting, and the compressed rule.
The raw model already has high AUC, but the exact-lag F1 is very low, around 0.048.
That means prediction can look good while attribution is poor.
After rectification, AUC improves to about 0.998, Youden's J improves, exact-lag F1 rises to about 0.619, and pairwise support stability also improves.
The compressed rule preserves most of the discrimination while reducing the active support to about six terms on average and improving exact-lag F1 further.
So the main synthetic result is not just better prediction. It is better attribution, stability, compactness, and efficiency.

# RQ1: Quadratic Programming Comparison




- Katrutsa–Strijov QP provides feature-importance weights, but not a full predictive model for AUC, Youden's J, or F1 evaluation. For comparison, an L1 fit was applied to the selected feature set.
- Many importance values came close to identifying the correct variables, but in several cases selected an incorrect lag.

| Method | AUC | Youden's J-index | Exact-lag F1 |
|---|---|---|---|
| QP + L1 | 0.894 | 0.603 | **0.000** |
| Rectified L1 | 0.997 | 0.946 | **0.571** |

**Both methods compared on a level playing field**

The original Katrutsa–Strijov implementation was in R and ran much slower. For a level comparison, the method was reimplemented in Python; after reimplementation, runtime was comparable.

SCRIPT
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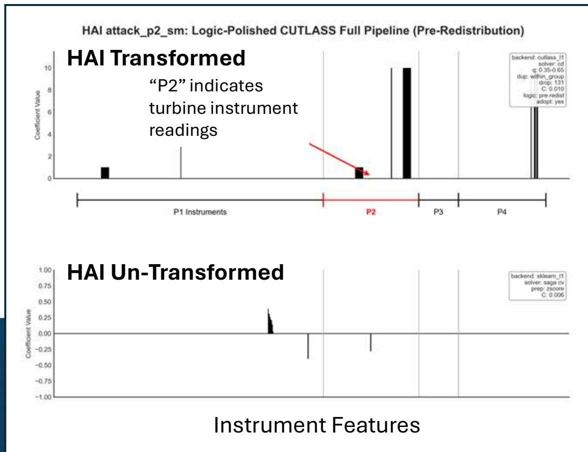
This slide compares rectification with a close parallel method in intent: the Katrutsa-Strijov quadratic-programming feature-selection approach.
The QP method provides feature-importance weights, but it does not directly produce a predictive model for metrics like AUC, Youden's J, or F1. So for comparison, an L1 fit was applied to the selected feature set.
The QP importance values often came close to identifying the correct variables, but in several cases they selected an incorrect lag.
The table shows the key result. QP plus L1 had AUC of 0.894, Youden's J of 0.603, and exact-lag F1 of 0.000. Rectified L1 had AUC of 0.997, Youden's J of 0.946, and exact-lag F1 of 0.571.
The bottom note addresses runtime fairness: the original QP implementation was in R, so it was reimplemented in Python. After that, runtime was comparable, so the difference here is not just an implementation artifact.

# RQ1: Real World Data

## HAI Exemplar and Cross-Domain Check




| Statistic | HAI (Rectified) | HAI (Raw) | UNICEF** (Rectified) | UNICEF** (Raw) |
|---|---|---|---|---|
| ACC | 0.987 | 0.939 | 0.902 | 0.963 |
| AUC | 0.982 | 0.943 | 0.865 | 0.989 |
| F1 | 0.987 | 0.942 | 0.894 | 0.958 |
| Youden's J-Index | 0.974 | 0.878 | 0.815 | 0.925 |

** UNICEF scope note: raw metrics were higher, but the raw model used a less compact feature set (16 vs. 5 categories), illustrating that rectification is not uniformly superior and must be judged against attribution and complexity goals.

### Rectified vs. Raw Feature Attribution

In the HAI exemplar, rectification concentrated coefficient mass on the turbine-loop P2 instruments while suppressing unrelated features. A smaller P4 response is plausible because P4 is the HIL simulator block and contains signals coupled to the P2 turbine-loop response.

SCRIPT

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This slide shows the real-world evidence using HAI as the main exemplar and UNICEF as a cross-domain check.

In the HAI case, rectification concentrated coefficient mass on the turbine-loop P2 instruments while suppressing unrelated features. That is exactly the kind of attribution behavior we want.

The smaller P4 response is plausible because P4 is the HIL simulator block and contains signals coupled to the P2 turbine-loop response.

The table also includes UNICEF. There, raw metrics were higher, but the raw model used a less compact feature set: 16 categories versus 5. That is an important scope point.

So the conclusion is not that rectification is universally superior on every metric. The conclusion is that rectification must be judged against attribution, compactness, and complexity goals, not only raw discrimination.

# RQ1 Answer

RQ1: “Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?”

The answer is **yes, conditionally**. Critical-range rectification improved support concentration, lag localization, compactness, and often runtime in threshold-and-lag aligned regimes. Cross-domain checks show that it should be used when threshold structure is meaningful and held-out performance is preserved.

### Transition to RQ2:

- *What mechanism explains the empirical gains?*
- *Under what assumptions can rectification improve sparse recovery?*
- *Where do the theoretical and empirical boundaries appear?*

### Efficiency Note

In the synthetic studies, rectification reduced runtime even after transformation overhead, while producing more compact and stable sparse supports.

SCRIPT
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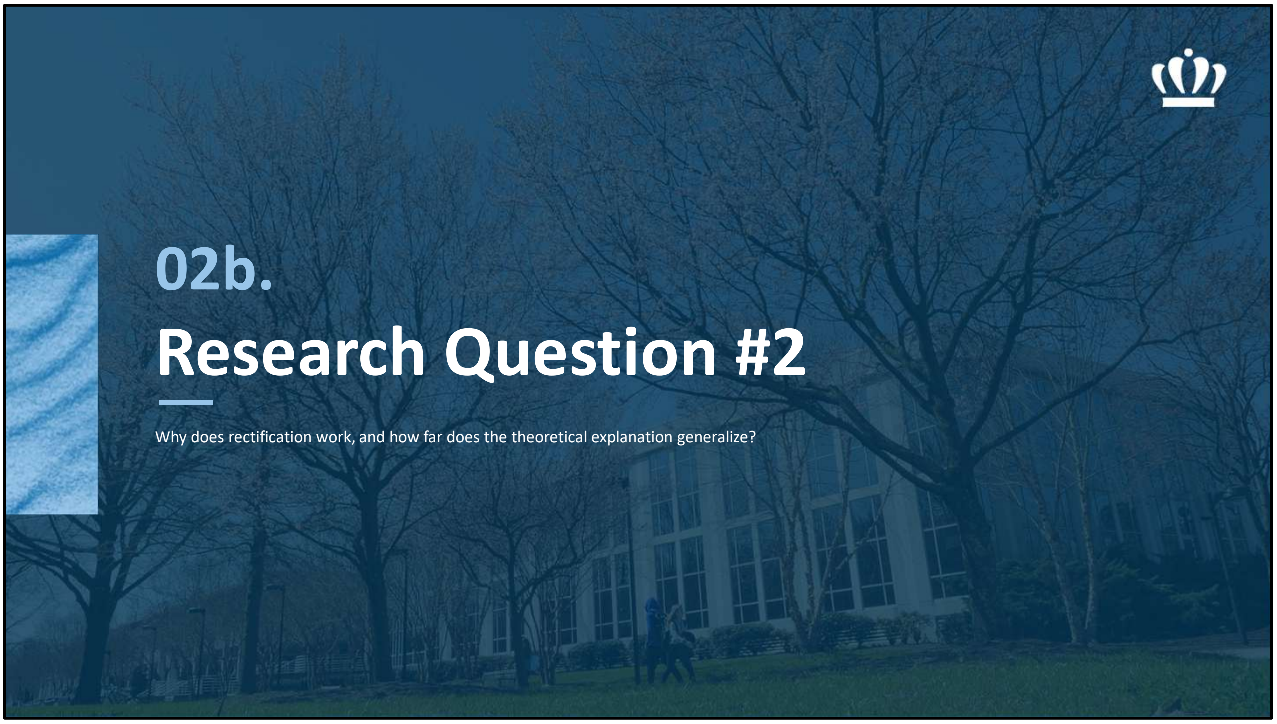
The answer to RQ1 is **yes, conditionally**.
Critical-range rectification improved support concentration, lag localization, compactness, and often runtime in threshold-and-lag aligned regimes.
The cross-domain checks show that it should be used when threshold structure is meaningful and held-out performance is preserved.
The efficiency note is also important. In the synthetic studies, rectification reduced runtime even after transformation overhead, while producing more compact and stable sparse supports.
This leads naturally to RQ2. The empirical results are strong in the target regime, but we still need to ask what mechanism explains the gains, under what assumptions rectification improves sparse recovery, and where the theoretical and empirical boundaries appear.




SCRIPT
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The second research question is:
**Why does rectification work, and how far does the theoretical explanation generalize?**
This section gives a scoped theoretical explanation. The goal is not to prove that every possible critical-range rectifier always helps.
Instead, the goal is to show a tractable mechanism: under explicit assumptions, a threshold transformation can contract dependence and improve sparse-recovery conditions.

# RQ2: Irrepresentable Condition (IC)

LASSO support recovery is favored when inactive features cannot mimic the active set:

$$\Big\|\overset{\text{Inact/Act Covariance}}{X_{S^c}^{\top}X_S}\,\overset{\text{Inverse Gram matrix}}{(X_S^{\top}X_S)^{-1}}\,\overset{\text{Sign vector}}{\mathrm{sign}(\beta_S)}\Big\|_{\infty} < 1$$

may become easier to satisfy after zero-threshold sign binarization under the theorem assumptions.

- Zhao and Yu's IC gives a formal lens for when LASSO can recover the true sparse support.
- Raw lag-expanded longitudinal features often violate this condition through high collinearity.
- RQ2 uses zero-threshold sign binarization as a tractable special case to show how rectification can contract dependence and improve IC behavior.
- The result is mechanistic and scoped, not a universal guarantee for every rectifier or dataset.

**Scoped Theoretical Mechanism**

The theorem analyzes zero-threshold sign binarization under explicit assumptions because that case admits a closed-form arcsin correlation relation. Practical critical-range rectification is broader, so later boundary checks test how far the mechanism appears to extend.

SCRIPT
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The theoretical lens we will use is the LASSO irrepresentable condition.
The IC gives a formal way to reason about when LASSO can recover the true sparse support. In high-dimensional lag-expanded longitudinal data, raw features often violate this condition because of high collinearity.
RQ2 uses zero-threshold sign binarization as a tractable special case. In that setting, we can show how rectification can contract dependence and improve IC behavior.
The important qualifier is that this is a scoped, mechanistic result. It is not a universal guarantee for every rectifier or dataset.
The bottom box captures the scope: practical critical-range rectification is broader, so the later boundary checks test how far this mechanism appears to extend.

# RQ2: Parsing the Results

Taking the original IC expression:

$$\|X_{S^c}^{\top} X_S \, (X_S^{\top} X_S)^{-1} \mathrm{sign}(\beta_S)\|_{\infty} < 1$$

and defining a similar form:

$$\theta_j \;=\; C_{jS} \, G_{SS}^{-1} \, \mathrm{sign}\big(\beta_S\big)$$

then applying the submultiplicative property of induced norms:

$$|\theta_j| \le \|C_{jS}\|_{\infty} \|G_{SS}^{-1}\|_{\infty} \|sign(\beta_S)\|_{\infty}$$

- The sign vector $sign(\beta_S)$ has infinity norm 1 on the active support.
- Positive correlations are attenuated after zero-threshold sign binarization, reducing irrelevant correlations with the active set.
- Under the scoped assumptions, inactive-active covariance terms $(X_{S^c}^{T} X_S)$ shrink.
- For the scoped positive equicorrelation case, lower active-set correlation reduces the relevant $\| \, G_{SS}^{-1} \, \|_{\infty}$ bound, so the active-set inverse contributes less amplification to the IC term.

**Parsing the IC term**

The proof decomposes the IC term into three factors: inactive-active leakage, active-set inverse amplification (the inverse Gram matrix), and coefficient-sign direction. Sign-binarization reduces the first two under the scoped assumptions, making the IC bound easier to satisfy.

SCRIPT

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This slide decomposes the IC term.
The IC expression combines three components: inactive-active covariance, inverse active-set covariance, and the sign direction of the active coefficients.
The sign vector has infinity norm 1 on the active support, so the important changes occur in the covariance terms.
The arcsin relation attenuates positive correlations after zero-threshold sign binarization. Under the scoped assumptions, inactive-active covariance terms shrink.
That means inactive lag features are less able to imitate active lag features in the IC expression.
So this slide explains the mechanism: rectification can make sparse recovery easier because it reduces the dependence pathways that allow irrelevant features to masquerade as relevant ones.

NOTES

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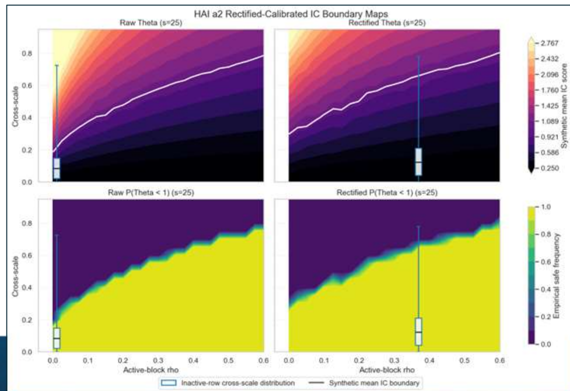
- Uses the Sherman-Morrisson-Woodbury formula for the inverse of a matrix.
- Under symmetric binarization, the binarized Gram matrix remains equicorrelated with effective correlation $\tilde{\rho}$.

- The resulting inequality follows from the closed-form row-sum expression of $G^{-1}$under equicorrelation.
- This slide decomposes the IC term into covariance between inactive and active features, inverse active-set covariance, and the signs of the active coefficients. The previous slides showed why the middle and covariance pieces can shrink after sign binarization.

# RQ2: Boundary Checks Beyond the Theorem




- Using the same IC-boundary framework as the theorem, I overlaid the HAI real-world data before and after rectification.
- Rectification shifts the empirical distribution into a more favorable region and increases contrast between active and inactive dependence structure.

| Boundary Check | Empirical Result | Meaning |
|---|---|---|
| Positive zero-threshold sign case | Strong | Clean match to theorem assumptions |
| Shifted one-sided cutoff | Strong | Practical thresholds can preserve the mechanism |
| Shifted interval rule | Weak | Broader rectifiers need empirical validation |
| Negative-side IC sweep | Did not hold | Not all correlation structures benefit |
| Negative pair + complement feature | Recovered | Representation design can repair some failures |

**RQ2 takeaway**

The theorem gives a scoped mechanism, not a universal guarantee. Rectification is strongest when the transformation contracts harmful dependence while preserving the event logic, and boundary checks can identify when caution or hybrid features are needed.

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This slide is important because it states the limits of the theory.
The theorem gives a scoped mechanism, not a universal guarantee. To test how far the mechanism extends, I used empirical boundary checks.
The contour plots show the HAI real-world data overlaid on the same kind of IC-boundary framework used in the theorem. Before and after rectification, we can see how the empirical distribution shifts into a more favorable region and increases contrast between active and inactive dependence structure.
The table summarizes the broader boundary package.
The positive zero-threshold sign case is strong and matches the theorem assumptions.
A shifted one-sided cutoff is also strong, suggesting practical thresholds can preserve the mechanism.
Shifted interval rules are weaker, meaning broader rectifiers need empirical validation.
The negative-side IC sweep did not hold, showing that not all correlation structures benefit.
And the negative pair with a complement feature shows that representation design can repair some failures.
The takeaway is that RQ2 provides a defensible scoped mechanism, while the

boundary checks define when caution or hybrid features are needed.

NOTES
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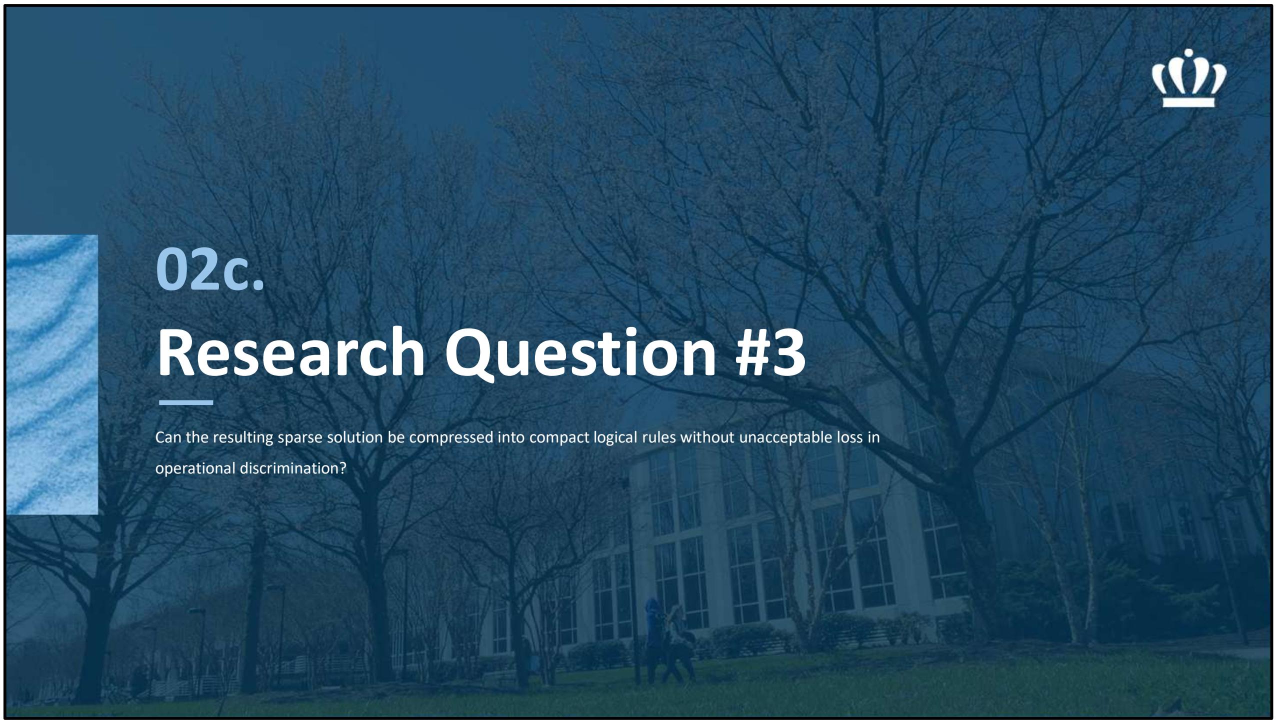
- The table summarizes the broader boundary checks.
- The highlighted row is the closest practical analogue for HAI-style critical-range rectification, which is broader than the clean zero-threshold theorem case.

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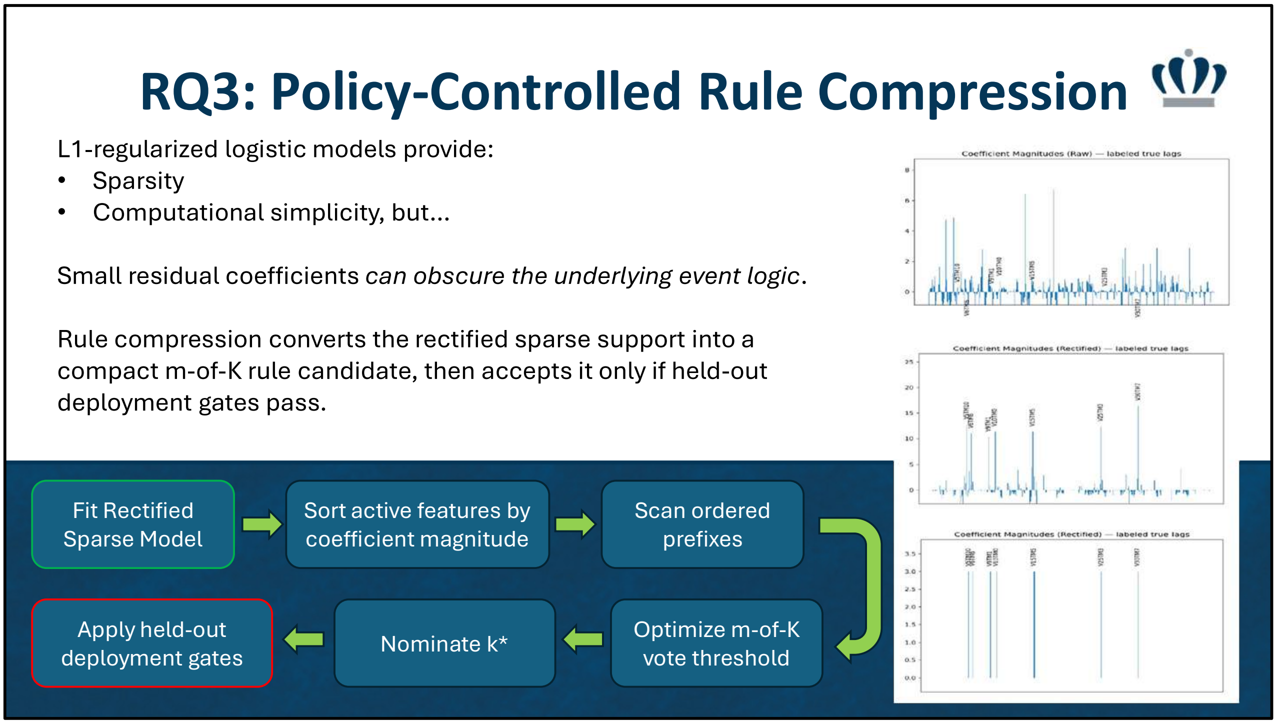
The third research question is:
**Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?**
This question moves from sparse attribution to deployment usability.
Even a sparse model may still have too many small coefficients for direct human use.
RQ3 asks whether the rectified sparse model can be turned into a simpler rule artifact, and whether we can decide when that simplification is safe.

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L1-regularized logistic models provide sparsity and computational simplicity, but they can still leave small residual coefficients that obscure the underlying event logic.
Rule compression converts the rectified sparse support into a compact m-of-K rule candidate. The key word is **candidate**.
The flow diagram shows the process.
First, fit the rectified sparse model. Then sort active features by coefficient magnitude. Then scan ordered prefixes. For each prefix, optimize the m-of-K vote threshold. Then nominate a candidate k-star.
Finally, the candidate is accepted only if held-out deployment gates pass.
So compression is not a post-hoc simplification chosen by preference. It is a policy-controlled procedure with an explicit accept-or-reject decision.

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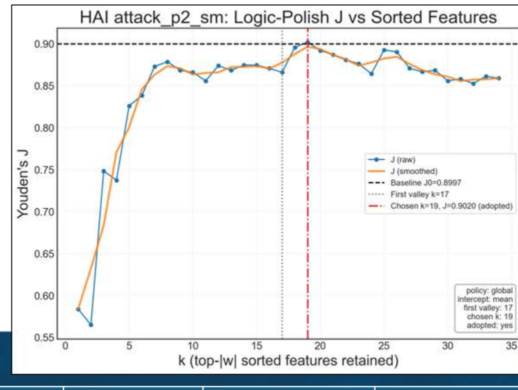
The compression stage is not meant to replace the sparse model automatically. It searches for a compact m-of-K rule candidate over the active rectified support, then accepts the rule only if held-out deployment gates show no unacceptable loss in operational discrimination.

# RQ3: HAI Compression Policy Results

The Youden's J vs K plot shows the number of features required to achieve each specific result for the HAI dataset.

- The anytime frontier scans ordered prefixes of active features and identifies compact m-of-K rule candidates.
- Compression is adopted only when the candidate passes held-out deployment gates; otherwise, the rectified sparse baseline is retained.
- In HAI, the policy selected compressed rules for P2 and P1/P2, but rejected candidates for P3 and P1/P3.




### HAI Real World Data Test

| Domain | Baseline nz | Candidate k* | Compression | $\Delta AUC$ | $\Delta J$ | Deploy? |
|---|---|---|---|---|---|---|
| HAI Attack P2 | 39 | 14 | 2.79x | -0.011 | +0.015 | Yes |
| HAI Attack P1 & P2 | 25 | 8 | 3.12x | -0.001 | -0.008 | Yes |
| HAI Attack P3 | 23 | 7 | 3.29x | -0.003 | +0.341 | No* |
| HAI Attack P1 & P3 | 34 | 8 | 4.25x | +0.011 | +0.014 | No |

SCRIPT

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This slide shows the HAI compression policy results.

The Youden's J versus K plot shows the anytime frontier: as we retain more features, performance changes along the ordered feature path.

The frontier identifies compact m-of-K rule candidates, but compression is adopted only when the candidate passes the held-out deployment gates.

The table shows the result.

For HAI Attack P2 and HAI Attack P1/P2, the policy selected compressed rules. These give roughly 2.8x and 3.1x compression.

For HAI Attack P3 and HAI Attack P1/P3, the policy rejected the candidates, so the upstream rectified sparse baseline is retained.

P3 is a useful example of why the policy gate matters. Even though the candidate improved Youden's J, its 95% confidence interval exceeded the allowed delta-AUC gate, so the compressed rule was rejected. The deployment decision is governed by the full held-out policy, not by a single improved metric.

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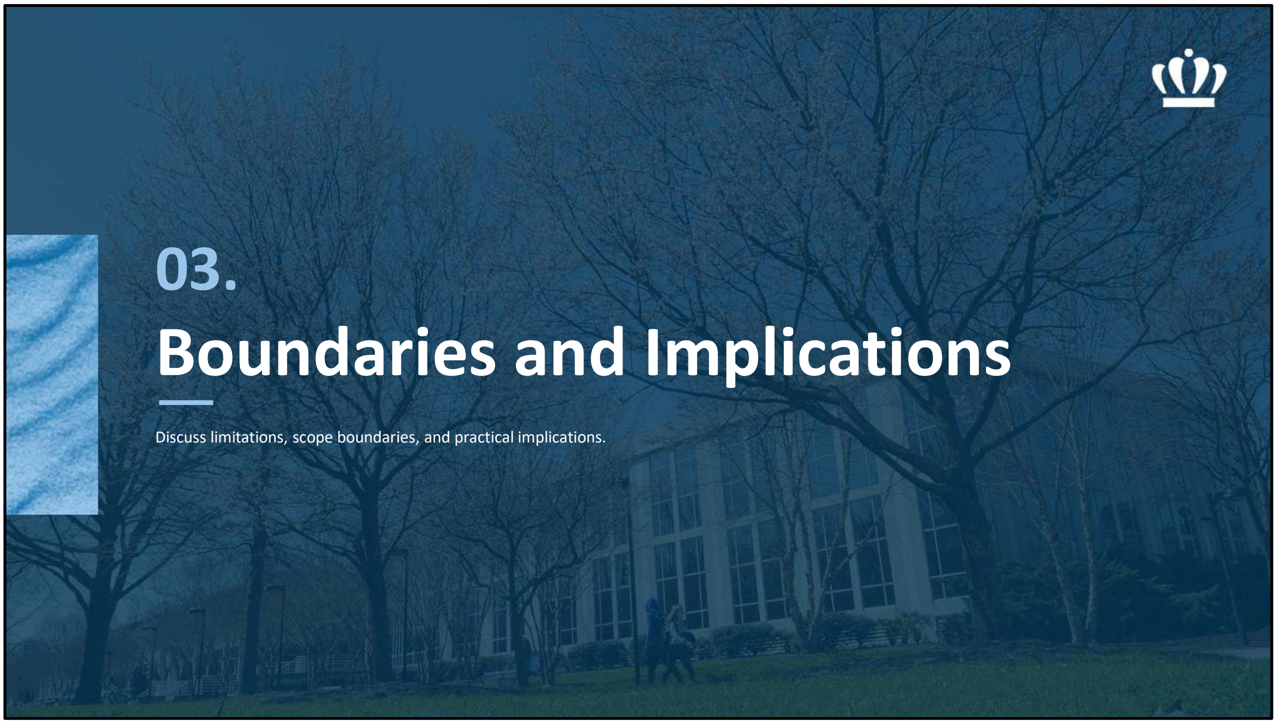
P3 is a useful example of why the policy gate matters. Even though the candidate improved Youden's J, its 95% confidence interval exceeded the allowed ΔAUC gate, so the compressed rule was rejected and the upstream sparse baseline was retained. This illustrates that the deployment decision is governed by the full held-out policy, not by a single improved metric.

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I will now step back from the individual research questions and discuss the boundaries and practical implications.
This section is important because the dissertation does not claim that rectification and compression are universally appropriate.
Instead, the contribution is a pipeline with diagnostics, gates, and fallback decisions.

# Scope Boundaries

**Best-fit regime**

- Threshold-mediated event structure
- Meaningful lagged dependencies
- Raw sparse models remain predictive but unstable
- Rectification improves attribution, compactness, or stability without harming held-out fit

**Use caution when**

- Threshold structure is weak or unstable
- Small threshold changes produce erratic results
- Post-threshold magnitude still carries important signal
- Compression fails held-out non-inferiority gates

**Takeaway**

Rectification is a representation strategy, not a universal replacement for raw continuous modeling.

SCRIPT
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This slide summarizes when the method is most appropriate and when caution is needed.
The best-fit regime has threshold-mediated event structure, meaningful lagged dependencies, and raw sparse models that remain predictive but unstable.
Rectification is especially useful when it improves attribution, compactness, or stability without harming held-out fit.
Caution is needed when threshold structure is weak or unstable, when small threshold changes produce erratic results, when post-threshold magnitude still carries important signal, or when compression fails held-out non-inferiority gates.
The takeaway is that rectification is a representation strategy, not a universal replacement for raw continuous modeling.
The correct decision may be to stay raw, use hybrid features, or stop before compression.

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The method is strongest when the problem is genuinely threshold-and-lag aligned.
When the data still depend heavily on smooth magnitude effects, or when threshold

diagnostics are unstable, the correct decision is to stay raw, use hybrid features, or stop before compression.

# Practical Implications

## A Deployment Playbook

### How to use the pipeline

1. Start with a raw sparse baseline.
2. Screen for threshold-and-lag structure.
3. Estimate critical ranges on training data only.
4. Compare raw vs. rectified attribution and held-out discrimination.
5. Compress only when a simpler rule is needed and justified.
6. Deploy compressed rules only when held-out gates pass.

### What the user gets

- More stable feature-lag attribution.
- Smaller, more auditable sparse supports.
- Optional compact m-of-K rules.
- Explicit fallback decisions when simplification is unsafe or unnecessary.

**Takeaway**

The contribution is not just a model; it is a decision policy for interpretable longitudinal modeling.

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This slide translates the dissertation into a deployment playbook.
The workflow is simple.
Start with a raw sparse baseline. Screen for threshold-and-lag structure. Estimate critical ranges on training data only. Compare raw and rectified attribution and held-out discrimination. Compress only when a simpler rule is needed and justified. Deploy compressed rules only when held-out gates pass.
What the user gets is more stable feature-lag attribution, smaller and more auditable sparse supports, optional compact m-of-K rules, and explicit fallback decisions when simplification is unsafe or unnecessary.
The practical contribution is not just a model. It is a decision policy for interpretable longitudinal modeling.

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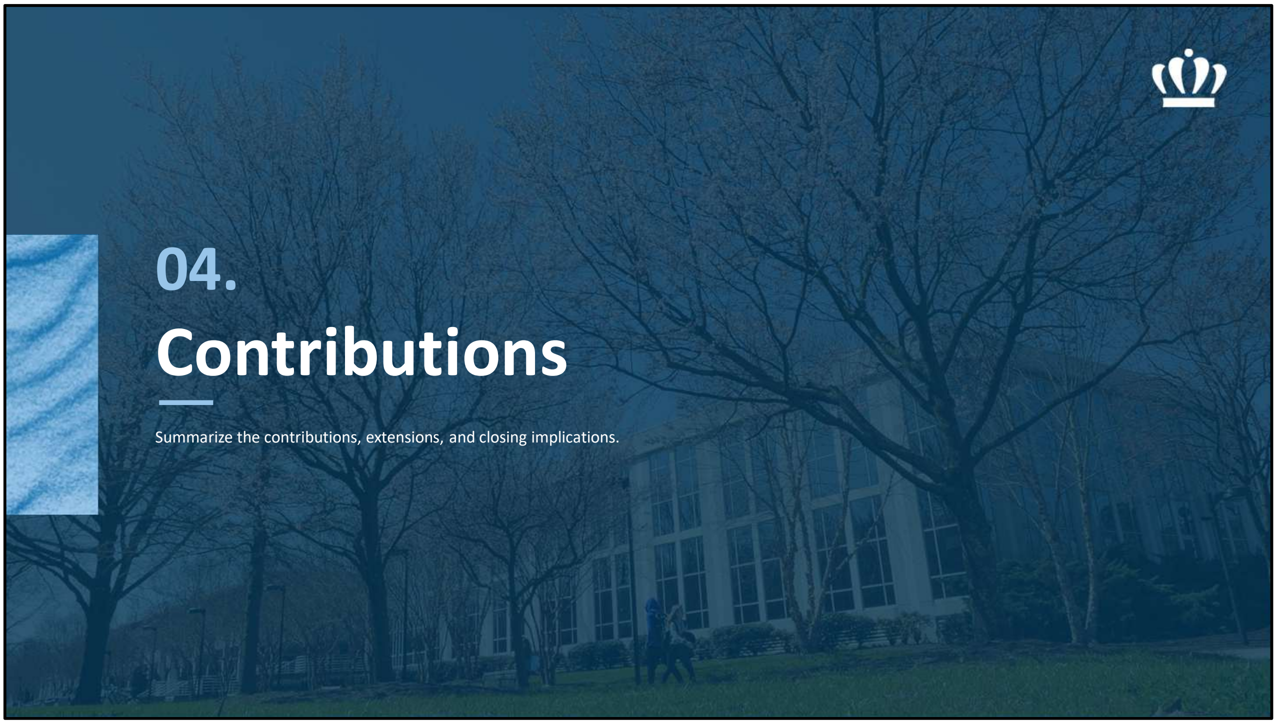
The practical implication is that this framework gives analysts a structured way to decide whether to use raw sparse modeling, rectified sparse modeling, or a compressed rule. The important point is that simplification is governed by evidence, not by preference for simpler models alone.




SCRIPT
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I will close by summarizing the dissertation contributions.
The key point is that the work is not just a collection of experiments. It contributes an integrated framework: a representation-first pipeline, a scoped theoretical bridge, an anytime compression method, an empirical validation package, and a reproducible implementation pathway.

# Contributions

1. **Unified rectification-first pipeline**
   - Complete workflow from raw longitudinal data to sparse models and optional rules.
2. **Scoped theoretical bridge**
   - Mechanistic explanation for why rectification can improve sparse recovery under explicit assumptions.
3. **Anytime rule compression stage**
   - Converts active rectified support into compact m-of-K rule candidates when policy gates justify simplification.
4. **Empirical validation package**
   - Synthetic, benchmark, HAI, UNICEF, ablation, and boundary studies characterize both gains and limits.
5. **Reproducible implementation pathway**
   - CUTLASS package and notebooks connect the dissertation artifacts to reusable analyses.

**Integrated Research Artifact**

The dissertation contributes both a theoretical framing and a usable implementation pathway for interpretable longitudinal modeling.

SCRIPT
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The dissertation makes five main contributions.
First, it provides a unified rectification-first pipeline: a complete workflow from raw longitudinal data to sparse models and optional rules.
Second, it provides a scoped theoretical bridge: a mechanistic explanation for why rectification can improve sparse recovery under explicit assumptions.
Third, it provides an anytime rule-compression stage that converts active rectified support into compact m-of-K rule candidates when policy gates justify simplification.
Fourth, it provides an empirical validation package, including synthetic studies, benchmark comparisons, HAI, UNICEF, ablation, and boundary studies that characterize both gains and limits.
Fifth, it provides a reproducible implementation pathway through the CUTLASS package and notebooks that connect the dissertation artifacts to reusable analyses.
Taken together, the dissertation contributes both a theoretical framing and a usable implementation pathway for interpretable longitudinal modeling.


SCRIPT
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Thank you.
The final message is that critical-range rectification can improve sparse longitudinal attribution when the event structure is threshold-and-lag aligned; the theoretical analysis provides a scoped mechanism for why this can happen; and anytime rule compression makes the result operationally usable when held-out policy gates justify simplification.
I’m happy to take questions.

OLD DOMINION UNIVERSITY

# BACKUP SLIDES

# RQ2: The Arcsin Relation

The bivariate normal orthant probability identity, $P(X > 0, Y > 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho)$ implies the following closed-form mapping:

$$\tilde{\rho} \;=\; \frac{2}{\pi}\arcsin(\rho),$$

hence $|\tilde{\rho}| \leq |\rho|$ for $\rho \in (-1{,}1)$

- For standardized, jointly normal variables with correlation $\rho$, zero-threshold sign binarization induces the closed-form arcsin mapping.
- The tilde ("~") marks the binarized feature correlation.
- Under these assumptions, binarized pairwise correlations satisfy $|\tilde{\rho}| \leq |\rho|$.
- **Equality occurs only at $\rho \in \{-1, 0, 1\}$**; otherwise, the transformed correlation is smaller in magnitude.

**Key intuition**

Sign binarization contracts pairwise correlations, reducing harmful dependence among lag-expanded features.

SCRIPT
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The key mathematical tool is the arcsin relation.
For standardized, jointly normal variables with correlation rho, zero-threshold sign binarization induces a closed-form mapping.
The transformed correlation is:

$$\rho_{tilde} = (2/pi) * arcsin(\rho)$$

Under these assumptions, the magnitude of the transformed pairwise correlation is no larger than the original correlation.
The important intuition is at the bottom: **sign binarization contracts pairwise correlations**, and that reduces harmful dependence among lag-expanded features.
This is the bridge from the representation change to improved sparse-selection behavior.

NOTES
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In the Gaussian/elliptical case, the correlation structure induced by sign-binarization is exactly Kendall's tau. Thus, sign-binarization (as presented) is effectively converting linear dependence into monotone / ordinal dependence. This also might imply less sensitivity to outliers or heavy tails, making the solutions more robust. However, we

have not studied these aspects, yet.

# RQ2: Gram Matrix Bounds

The Gram matrix $X_S^T X_S$ is expressed as $G$ and its inverse as $G^{-1}$ in the following equicorrelated case assuming $\rho \in (-1/(s-1), 1)$ so that $G$ is invertible:

$$G = n\big((1-\rho)I_s + \rho J_s\big),$$

$$G^{-1} = \frac{1}{n(1-\rho)}\left(I_s - \frac{\rho}{1+\rho(s-1)}J_s\right)$$

where, $I_S$ is the identity matrix of size $S$, and $J_S$ is a matrix of ones of size $S$.

- Similar expressions for the binarized versions $\tilde{G}$ and $\tilde{G}^{-1}$ and can be obtained by substituting $\tilde{\rho}$ for $\rho$.
- When the expression $0 < \tilde{\rho} < \rho < 1$ is true, it can be shown that:

$$\| \tilde{G}^{-1} \|_\infty \leq \| G^{-1} \|_\infty$$

- Thus, sign binarization contracts positive correlations in this equicorrelated setting, reducing the inverse-Gram contribution to the IC term. Recall that this is the second term in the IC:

$$\|X_{S^c}^\top X_S \, (X_S^\top X_S)^{-1} \mathrm{sign}(\beta_S)\|_\infty < 1$$

**Key bridge to IC**

The inverse Gram matrix is part of the IC term. In the scoped equicorrelated case, sign binarization contracts correlations and can reduce the inverse-Gram contribution, making support recovery more favorable.

SCRIPT
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The next step is to connect pairwise correlation contraction to the Gram matrix.
In the equicorrelated case, the active-set Gram matrix and its inverse have closed-form expressions. Similar expressions can be obtained for the binarized version by substituting the transformed correlation for the raw correlation.
When the transformed correlation is smaller, the inverse-Gram contribution can also be bounded.
The reason this matters is that the inverse Gram matrix appears directly inside the IC term. So the correlation contraction is not just an isolated fact about pairwise features. It affects the stability of the active-set inverse and the support-recovery condition.
The key bridge is that, in this scoped equicorrelated case, sign binarization can reduce the inverse-Gram contribution, making support recovery more favorable.

NOTES
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- Uses the Sherman-Morrisson-Woodbury formula for the inverse of a matrix.
- Under symmetric binarization, the binarized Gram matrix remains equicorrelated with effective correlation $\tilde{\rho}$.

- The resulting inequality follows from the closed-form row-sum expression of $G^{-1}$under equicorrelation.

# RQ2: Putting the Pieces Together

Since:

$$|\theta_j| \le \|C_{jS}\|_\infty \|G_{SS}^{-1}\|_\infty \|sign(\beta_S)\|_\infty$$

and:

$$\|\tilde{C}_{js}\|_\infty \le \|C_{js}\|_\infty,\ \|\tilde{G}_{ss}^{-1}\|_\infty \le \|G_{ss}^{-1}\|_\infty$$

we define:

$$B_j := \|C_{js}\|_\infty \|G_{ss}^{-1}\|_\infty,\ \tilde{B}_j := \|\tilde{C}_{js}\|_\infty \|\tilde{G}_{ss}^{-1}\|_\infty$$

$$B^* := max_{j \epsilon S^c}(B_j),\ \tilde{B}^* := max_{j \epsilon S^c}(\tilde{B}_j)$$

Thus, under the stated assumptions:

$$\Pr(\tilde{B}^* < 1) \ge \Pr(B^* < 1)$$

The lemmas isolate why binarization helps:

(i) pairwise contraction keeps off-diagonals from saturating
(ii) better conditioning bounds inverse norms
(iii) Smaller off-support covariances reduce the IC term.

Together these raise the chance of correct support recovery and promote sparse, stable selections.

**Scoped theoretical synthesis**

Under the stated assumptions, sign binarization contracts pairwise dependence, improves inverse-Gram behavior, and reduces inactive-active leakage in the IC term. This supports the mechanism for improved support recovery, while practical rectification remains broader and requires empirical boundary checks.

SCRIPT
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This backup slide shows the proof synthesis in one place.
The left side starts from the IC decomposition. The key terms are inactive-active covariance and the inverse active-set Gram matrix.
Under the stated assumptions, the previous lemmas show that sign binarization contracts the relevant covariance terms and improves the inverse-Gram behavior.
The result is a bound showing that the binarized version is more likely to satisfy the IC-related condition than the raw version in the theorem-scoped setting.
The important point is that the proof isolates three mechanisms: pairwise contraction, improved conditioning, and smaller off-support covariance.
Together, those mechanisms explain why rectification can promote sparse, stable selections under the scoped assumptions.

NOTES
=====
- Uses the Sherman-Morrisson-Woodbury formula for the inverse of a matrix.
- Under symmetric binarization, the binarized Gram matrix remains equicorrelated with effective correlation $\tilde{\rho}$.
- The resulting inequality follows from the closed-form row-sum expression of

$G^{-1}$under equicorrelation.

- In the Gaussian/elliptical case, the correlation structure induced by sign-binarization is exactly Kendall's tau. Thus, sign-binarization (as presented) is effectively converting linear dependence into monotone / ordinal dependence. This also might imply less sensitivity to outliers or heavy tails, making the solutions more robust. However, we have not studied these aspects, yet.